

NETWORKS AND PROTOCOLS

Lab 03 - Introduction to Wireshark

Introduction

In this first Wireshark lab, you'll get acquainted with Wireshark, and make some simple packet captures and observations.

The basic tool for observing the messages exchanged between executing protocol entities is called a packet sniffer. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a copy of packets that are sent/received from/by application and protocols executing on your machine.

Figure 1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or email client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure 1 is an addition to the usual software in your computer, and consists of two parts. The packet capture library receives a copy of every link-layer frame that is sent from or received by your computer over a given interface (link layer, such as Ethernet or WiFi). Recall from the discussion from section 1.5 in the text (Figure 1.24) that messages exchanged by higher layer protocols such as HTTP, FTP, TCP, UDP, DNS, or IP all are eventually encapsulated in link-layer frames that are transmitted over physical media such as an Ethernet cable or an 802.11 WiFi radio. Capturing all link-layer frames thus gives you all messages sent/received across the monitored link from/by all protocols and applications executing in your computer.

The second component of a packet sniffer is the packet analyzer, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. For example, suppose we are interested in displaying the various fields in messages exchanged by the HTTP protocol in Figure 1. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands

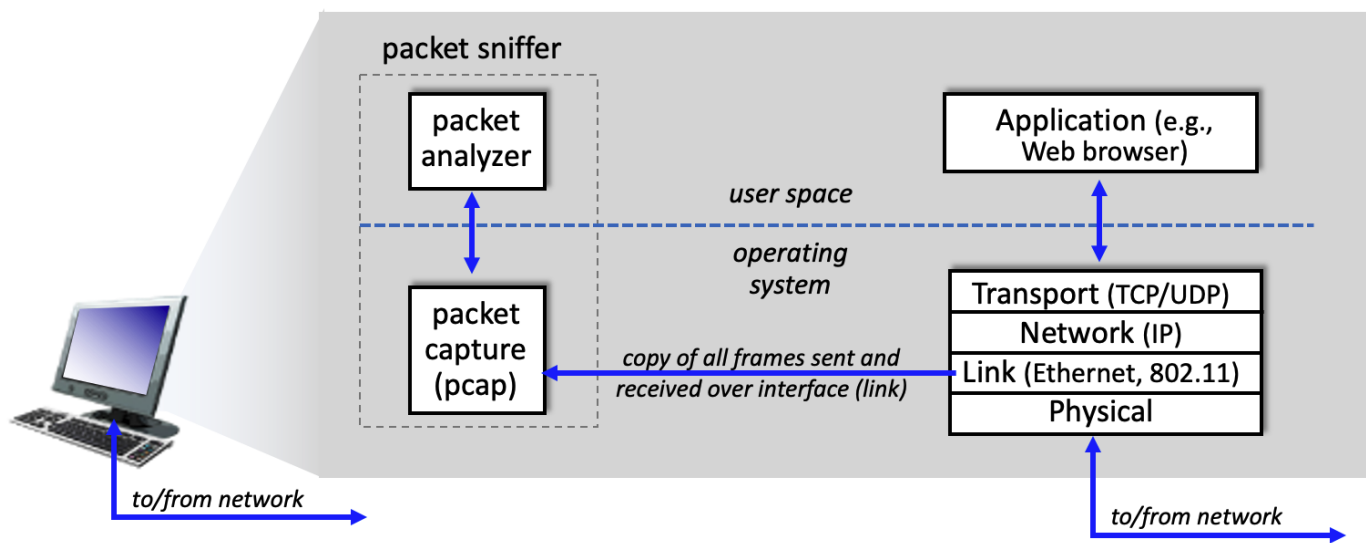


Figure 1: packet sniffer structure

the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

Getting Wireshark

In order to run Wireshark, you’ll need to have access to a computer that supports both Wireshark and the libpcap or WinPCap packet capture library. The libpcap software will be installed for you, if it is not installed within your operating system, when you install Wireshark. See <http://www.wireshark.org/download.html> for a list of supported operating systems and download sites.

Running Wireshark

When you run the Wireshark program, you’ll get a startup screen that looks something like the screen below. Different versions of Wireshark will have different startup screens – so don’t panic if yours doesn’t look exactly like the screen below!

There’s not much that’s very interesting on this screen. But note that under the Capture section, there is a list of so-called interfaces. The Mac computer we’re taking these screenshots from has just one interface – “Wi-Fi en0,” (shaded in blue in Figure 2) which is the interface for Wi-Fi access. All packets to/from this computer will pass through the Wi-Fi interface, so it’s here where we’ll want to capture packets. On a Mac, double click on this interface (or on another computer locate the interface on startup page through which you are getting Internet connectivity, e.g., mostly likely a WiFi or Ethernet interface, and select that interface).

Let’s take Wireshark out for a spin! If you click on one of these interfaces to start packet capture (i.e., for Wireshark to begin capturing all packets being sent to/from that interface), a screen like the one below will be displayed, showing information about the packets being captured. Once you start packet capture, you can stop it by using the Capture pull down menu

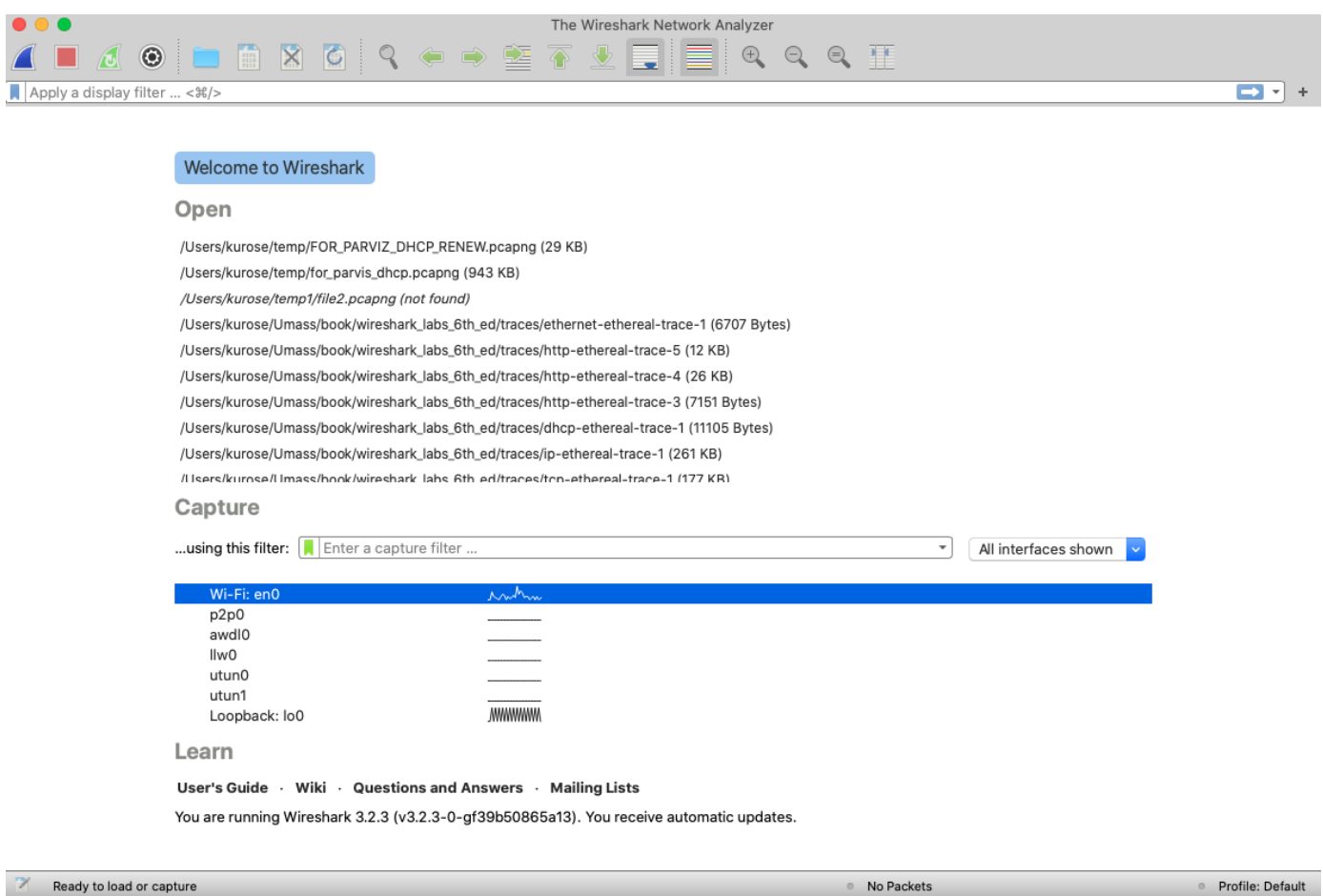


Figure 2: Initial Wireshark screen

and selecting Stop (or by clicking on the red square button next to the Wireshark fin in Figure 2).

The Wireshark interface has five major components:

- The **command menus** are standard pulldown menus located at the top of the Wireshark window (and on a Mac at the top of the screen as well; the screenshot in Figure 3 is from a Mac). Of interest to us now are the File and Capture menus. The File menu allows you to save captured packet data or open a file containing previously captured packet data and exit the Wireshark application. The Capture menu allows you to begin packet capture.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number (assigned by Wireshark; note that this is not a packet number contained in any protocol's header), the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet. The packet listing can be sorted according to any of these categories by clicking on a column name. The protocol type field lists the highest-level protocol that sent or received this packet, i.e., the protocol that is the source or ultimate sink for this packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. (To select a packet in the packet-listing window,

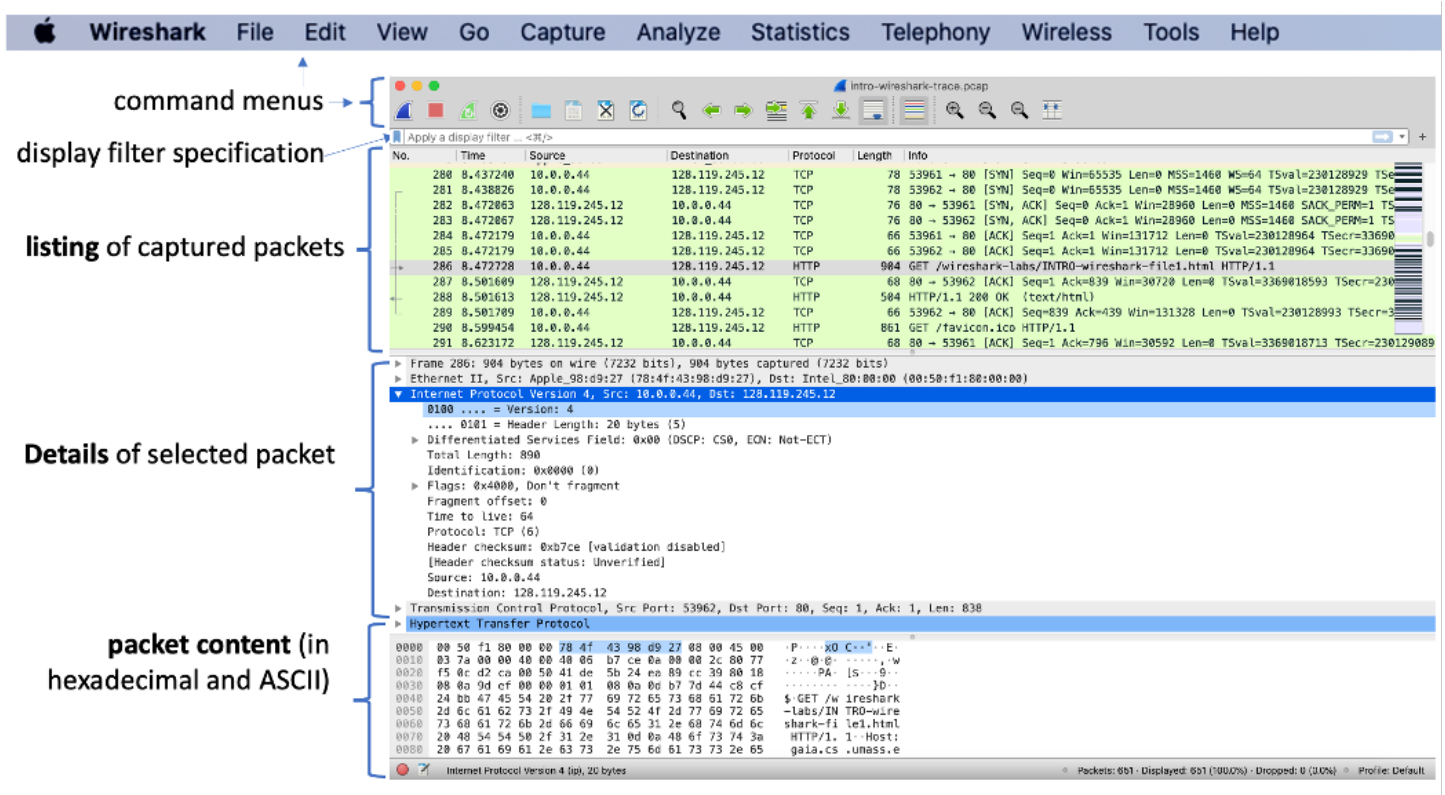


Figure 3: Wireshark window, during and after capture

place the cursor over the packet's one-line summary in the packet-listing window and click with the left mouse button.). These details include information about the Ethernet frame (assuming the packet was sent/received over an Ethernet interface) and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus/minus boxes or right/downward-pointing triangles to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.

- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.
- Towards the top of the Wireshark graphical user interface, is the **packet display** filter field, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window (and hence the packet-header and packet-contents windows). In the example below, we'll use the packet-display filter field to have Wireshark hide (not display) packets except those that correspond to HTTP messages.

Taking Wireshark for a Test Run

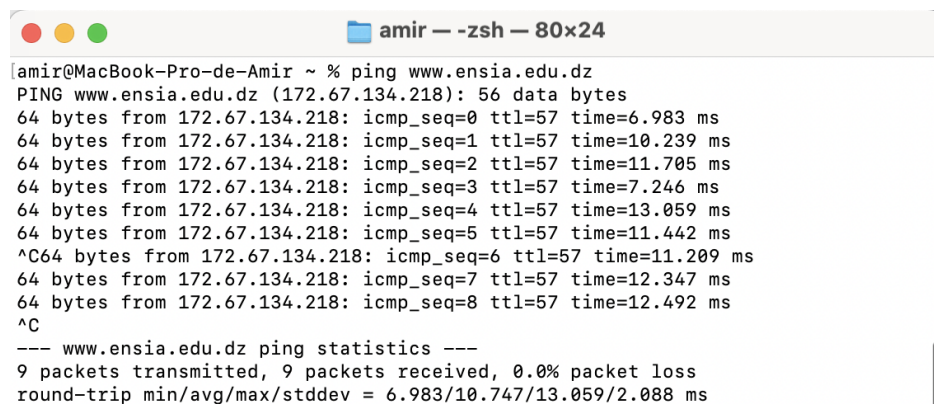
We'll assume that your computer is connected to the Internet via a wired Ethernet interface or a wireless 802.11 WiFi interface. Do the following:

1. Start up your favorite web browser, which will display your selected homepage.
2. Start up the Wireshark software. You will initially see a window similar to that shown in Figure 2. Wireshark has not yet begun capturing packets.
3. To begin packet capture, select the Capture pull down menu and select Interfaces. This will cause the “Wireshark: Capture Interfaces” window to be displayed (on a PC) or you can choose Options on a Mac.
4. Color Coding: You’ll probably see packets highlighted in green, blue, and black. Wireshark uses colors to help you identify the types of traffic at a glance. By default, green is TCP traffic, dark blue is DNS traffic, light blue is UDP traffic, and black identifies TCP packets with problems — for example, they could have been delivered out-of-order.
5. Once you begin packet capture, a window similar to that shown in Figure 3 will appear. This window shows the packets being captured. By selecting Capture pulldown menu and selecting Stop, or by click on the red Stop square, you can stop packet capture. But don’t stop packet capture yet. Let’s capture some interesting packets first. To do so, we’ll need to generate some network traffic.
6. Let’s do so using a web browser, which will use the HTTP protocol that we will study in detail in class to download content from a website.

Task 1: Capture the ENSIA website traffic

Step 1:

- We first need to know the IP address of ENSIA HTTP server. To do so, we need to ping the ENSIA website (figure 4). It appears that the IP address is 172.67.134.218



```
amir@MacBook-Pro-de-Amir ~ % ping www.ensia.edu.dz
PING www.ensia.edu.dz (172.67.134.218): 56 data bytes
64 bytes from 172.67.134.218: icmp_seq=0 ttl=57 time=6.983 ms
64 bytes from 172.67.134.218: icmp_seq=1 ttl=57 time=10.239 ms
64 bytes from 172.67.134.218: icmp_seq=2 ttl=57 time=11.705 ms
64 bytes from 172.67.134.218: icmp_seq=3 ttl=57 time=7.246 ms
64 bytes from 172.67.134.218: icmp_seq=4 ttl=57 time=13.059 ms
64 bytes from 172.67.134.218: icmp_seq=5 ttl=57 time=11.442 ms
^C64 bytes from 172.67.134.218: icmp_seq=6 ttl=57 time=11.209 ms
64 bytes from 172.67.134.218: icmp_seq=7 ttl=57 time=12.347 ms
64 bytes from 172.67.134.218: icmp_seq=8 ttl=57 time=12.492 ms
^C
--- www.ensia.edu.dz ping statistics ---
9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 6.983/10.747/13.059/2.088 ms
```

Figure 4: ENSIA ping result

- Now we have to launch packet sniffing with Wireshark
- After that, we launch our browser and we will open the ENSIA website (<https://ensia.edu.dz>)

- Now that the website is open, we will filter the sniffed frames on Wireshark using the filter bar. Use this syntax `ip.dst == 172.67.134.218` in that bar in order to display all the website packets. (Figure 5)

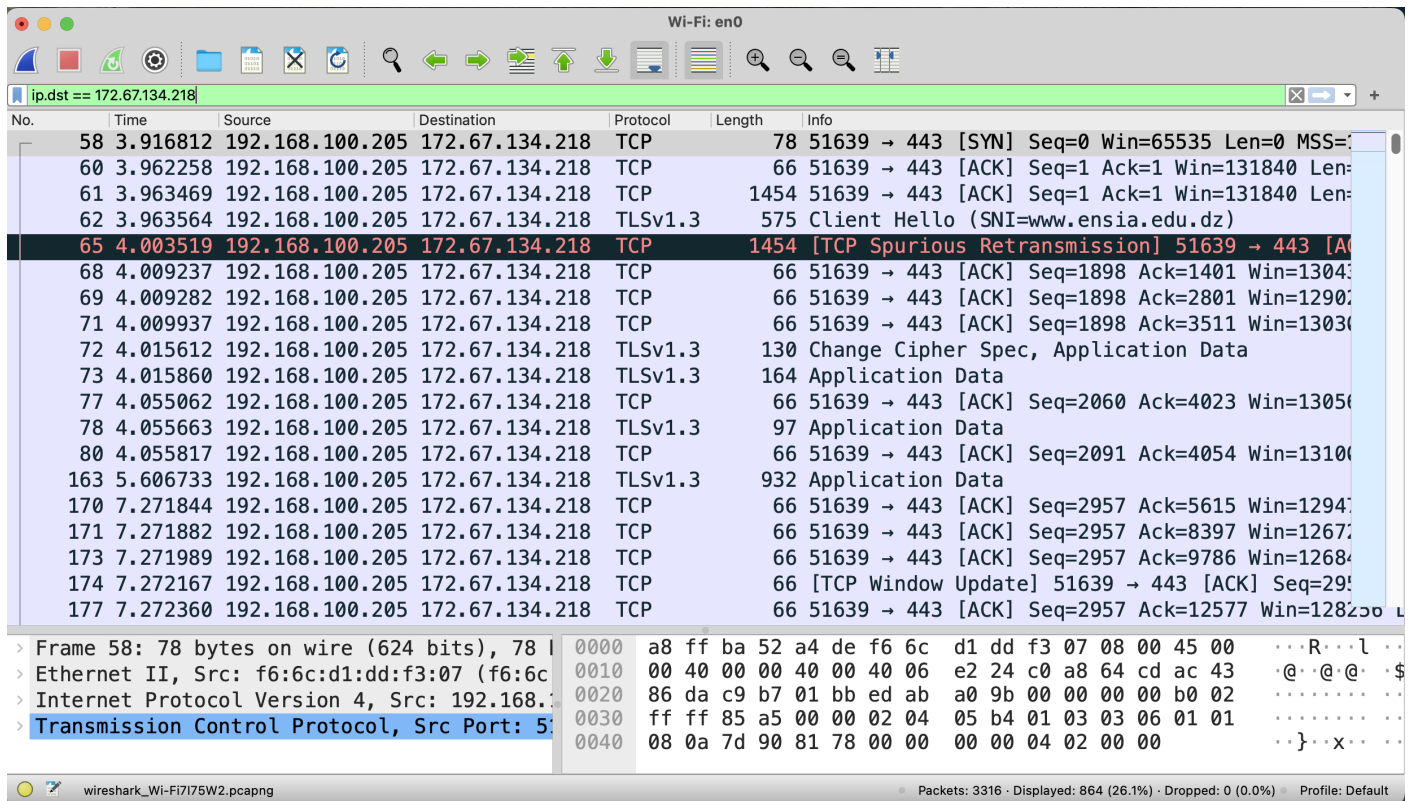


Figure 5: Wireshark using the IP filter